

Comparison of characteristics associated with super cyclonic storm ‘Amphan’ using numerical model WRF-ARW analysis and ERA5 reanalysis

Arun Kumar¹, Sushil Kumar^{2*}, Nagendra Kumar³, Nitin Lohan⁴ and R. Bhatla⁵

^{1,3}Department of Mathematics, M.M.H. College, Ghaziabad 201001, India.

^{1,2,4}Department of Applied Mathematics, Gautam Buddha University, Greater Noida 201312, India.

⁵Department of Geophysics, Institute of Science, Banaras Hindu University Varanasi 221005, India.

*Corresponding author: sushil.kumar@gbu.ac.in

ABSTRACT

The numerical simulation is performed to study the super cyclonic storm (SuCS) ‘Amphan’ by using the advanced research weather research and forecasting (WRF-ARW) model with 9 km resolution and in single domain while running WRF Pre Processing System (WPS) over the Bay of Bengal (BoB). The super cyclonic storm (SuCS) ‘Amphan’ originated over south Andaman Sea, which touched southeast BoB on 13th May 2020. The super cyclonic storm (SuCS) ‘Amphan’ underwent rapid intensification into a very severe cyclonic storm (VSCS) on 17th May 2020 and subsequently converted into extremely severe cyclonic storm (ESCS) on 18th May 2020 and finally into a super cyclonic storm (SuCS) around 1200 UTC of 18th May 2020 till next 24 hours, before weakening into ESCS and laid down as a low-pressure region over north Bangladesh and neighbourhood around mid-night of 21st May 2020. In this study, high resolution advanced research weather research and forecasting (WRF-ARW v4.1.2) with the combination of the Yonsei University (YSU) planetary boundary layer (PBL), Kain-Fritsch (KF) cumulus convection, and Ferrier microphysics scheme, is used for the simulation of SuCS ‘Amphan’. Spatial distribution of wind, potential vorticity (PV), and vertical integrated moisture transport (VIMT) are analysed for the intensity and the behaviour of the characteristics of SuCS ‘Amphan’. Fifth generation of European center for medium-range weather forecast (ECMWF) atmospheric reanalysis of global climate (ERA5) resolves the atmosphere with 137 levels from the surface at the height of 80 km. ERA5 gives hourly estimations of a big number of atmospheric variables and covers the Earth on a 31 km grid. ERA5 observations are used for the validation of WRF-ARW model results. It is observed that spatial distribution of wind and vertical integrated moisture transport (VIMT), are well matched with the observations. The study is a prelude for sensitivity analysis and data assimilation using four-dimensional variational data assimilation (4DVAR) technique.

Keywords: WRF-ARW Model, Tropical cyclone, Bay of Bengal, Super Cyclonic Storm (SuCS) Amphan, ERA5 Reanalysis.

INTRODUCTION

The Bay of Bengal (BoB) and Arabian sea (AS) are contributing almost seven percent of entire tropical cyclones (TCs) around the world (IMD Atlas, 2008). According to Indian Meteorological Department (IMD), tropical cyclones (TCs) over Bay of Bengal (BoB) are more frequently occurring than tropical cyclones (TCs) over Arabian sea (AS) (WMO, 2008). Every year tropical cyclones (TCs) are originating in North Indian Ocean (NIO) ranging in various categories and some of them are very much harmful for loss of life and economy of the larger area of Indian states. These tropical cyclones (TCs) are influenced by the bathymetry, and dynamical characteristics over North Indian Ocean (NIO), socio economic conditions, and dense population. The tropical cyclones (TCs), in general, are more frequent than landfalling tropical cyclones (TCs) over the Bay of Bengal. 8 cases of 10 highly severe tropical cyclones (TCs) that occurred in different part of world were from the Bay of Bengal (BoB) and Arabian sea (AS) (IMD Atlas, 2008). The simulation of the tropical cyclones (TCs) using mesoscale model is a necessary aspect to analysis of various parameters associated with the prediction of tropical cyclones (TCs). Therefore, the advanced research weather research and forecasting (WRF-ARW) model is used to study various parameters. The WRF-ARW model refers to the weather research and forecasting (WRF) model with the advanced

research weather research and forecasting (WRF) dynamical core. This version of the weather research and forecasting (WRF) model uses a more advanced set of equations and algorithms for atmospheric simulations, allowing for high-resolution and accurate weather forecasting. The comparison of the results WRF-ARW model and fifth generation of European center for medium-range weather forecast (ECMWF) atmospheric reanalysis of global climate (ERA5) reanalysis datasets are also shown in this study. Various global and mesoscale models e.g WRF-ARW model and, fifth generation of European center for medium-range weather forecast (ECMWF) atmospheric reanalysis of global climate (ERA5) reanalysis datasets, have been developed to improve at least 48 hours tropical cyclones TCs track prediction, intensity, and precipitations (Gupta, 2021). Many attempts have been made to improve performance of the model through nested domains, improving initial and boundary conditions (IBCs), and physical parameterizations, etc. Ooyama, (1969) simulated the life of tropical cyclones and discussed the basic mechanism of tropical cyclones (TCs). A cumulus parameterization (CP) scheme is studied to parametrize the convection in the models. Cumulus convection is an important factor in the progress and strengthening of tropical cyclones (TCs) (Anthes, 1977). Kain and Fritsch, (1990) have studied the dependency of cumulus parameterization (CP) scheme on the rate of sub grid

precipitation, latent heat release and redistribution of the thermodynamic and dynamic variables. Weisman et al., (1997) found that cumulus parameterization (CP) can be turned off if grid resolution is less than 4 km. They also showed that microphysics scheme can be allowed to resolve cloud and precipitation. In the last twenty-five years, it is observed that all weather forecasting activities are mostly dependent on numerical weather prediction (NWP) models to deliver progressive data of strength and their track due to changing TCs, statistical models and synoptical procedures (Mohanty and Gupta, 1997; Gupta, 2006; Chen et al., 2013; Mohapatra et al., 2013 a,b; Hegde et al., 2016; Routrey et al., 2019). Simulation of tropical cyclone 'Phalin' using WRF model was studied by Kumar et al., (2016). Singh et al., (2021) have studied an enormously severe cyclonic storm (SCS) over the BoB using WRF-ARW model.

Increasing model horizontal resolutions has been shown in several studies to have a favourable effect on tropical cyclone (TC) intensity forecast accuracy (Zhang et al., 2011; Gopalakrishnan et al., 2012). It is therefore anticipated that the forecast of tropical cyclone (TC) structure, and intensity will be improved by the use of a high-resolution modelling system. However, majority of these studies' initial and boundary conditions come from datasets like the Global Forecast System (GFS; $0.5^\circ \times 0.5^\circ$ resolution), ECMWF Reanalysis Interim/ERA5 ($0.25^\circ \times 0.25^\circ$ resolution), and NCEP Final analysis (FNL; $1^\circ \times 1^\circ$ resolution). The WRF-ARW model is available with different high-resolution. Therefore, it is critical to assess how well WRF-ARW model output data performs in forecasting severe tropical cyclones over the Bay of Bengal region.

Shenoy et al. (2021) have found that the best configuration for tropical cyclone forecasting over the Bay of Bengal is the combination of the Yonsei University (YSU) planetary boundary layer (PBL), Kain-Fritsch (KF) cumulus convection, and Ferrier microphysics scheme. The sensitivity studies show that the Kain-Fritsch (KF) cumulus parameterization yields the least errors in track (position and time) and intensity predictions when combined with the Yonsei University (YSU) planetary boundary layer (PBL) scheme and Ferrier (new eta) microphysics. Thus, the YSU-KF-Ferrier combo shows precise track and intensity forecasting (Shenoy et al., 2022). In this paper, an experiment is conducted using WRF-ARW model along with the combination of the Yonsei University (YSU) planetary boundary layer (PBL), Kain-Fritsch (KF) cumulus convection, and other fixed suitable options of model's configuration, for SuCS 'Amphan'. The main aim of this

paper is to analyse the wind, potential vorticity (PV), and vertical integrated moisture transport (VIMT) for the intensity and the behaviour of the characteristics of SuCS 'Amphan'. The Comparison of SuCS 'Amphan' using WRF model output data parameters with ERA5 model output parameters is also shown.

SUPER CYCLONIC STORM – 'AMPHAN'

SuCS 'Amphan' was shaped over the south Andaman Sea and adjoined south east BoB on 13th May 2020. It started concentrating into a deep depression (DD) over south east BoB on 16th May 2020. It further converted into a severe cyclonic storm (SCS) on 17th May 2020 morning, very severe cyclonic storm (VSCS) on 17th May 2020 afternoon, extremely severe cyclonic storm (ESCS) on 18th May 2020 morning and super cyclonic storm (SuCS) around afternoon of 18th May, 2020.

It intensified into a SuCS and SuCS maintained strength over west central Bay of Bengal (BoB) for almost 24 hours. Around afternoon of 19th May, SuCS started weakening in an ESCS over west central BoB. Therefore, ESCS weakened while crossing west Bengal-Bangladesh coasts as VSCS around Sunderbans during 1530 - 1730 UTC of 20th May with extreme nonstop wind speed 155-165 kmph, gusting to 185 kmph. It laid down over West Bengal as a VSCS on 20th May night. It was very near to Kolkata during this time interval. It kept on moving towards northeast and weakened into SCS over Bangladesh on 20th May mid night. It further weakened into CS on 21st May morning, converted into deep depression (DD) over Bangladesh around 21st May noon in the evening of the same day and into a depression (D) over northern Bangladesh. Around midnight on 21st May, 2020, it further weakened and laid down as a clearly defined low pressure region over north Bangladesh and the surrounding area.

DATA USED

The present study focuses over Bay of Bengal (BoB) region restricted between 78°E - 103°E longitude and 4°N - 26°N latitude and 9-km horizontal grid resolution (Figure 1). A single domain is used for the model simulation of length 126 hours of the above-mentioned region. The National Centres for Environmental Prediction (NCEP) operational analysis with $1^\circ \times 1^\circ$ horizontal resolution is used for initial conditions and boundary conditions to simulate the SuCS 'Amphan.' The Global Forecast System (GFS) is a numerical weather prediction model run by the NCEP. It provides global weather forecasts out to 16 days in advance, covering a wide range of meteorological parameters such as temperature, humidity, wind speed, and precipitation.

WPS Domain Configuration

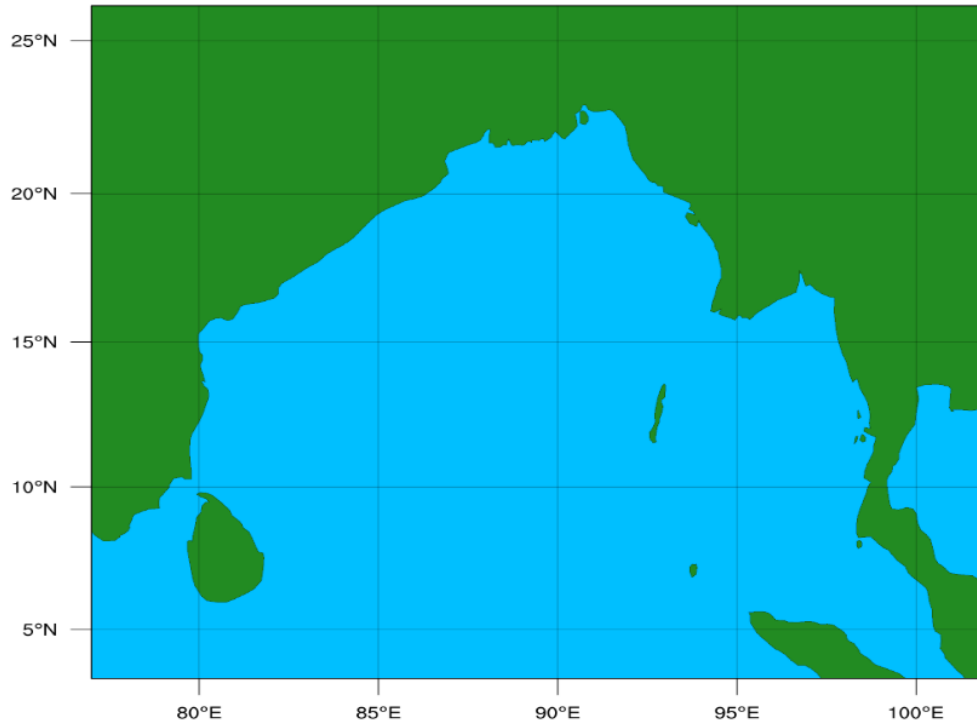


Figure 1. Advanced research weather research forecast (WRF-ARW) model domain for super cyclonic storm (SuCS) ‘Amphan’ study.

The lateral boundary conditions are downloaded from the NCEP (GFS) forecast fields at every 6 hours interval. ERA5 datasets are used for the validation of the parameters obtained from the model integration. The European Centre for Medium-Range Weather Forecasts (ECMWF) has developed a reanalysis dataset known as ERA5. It provides hourly data with high resolution (around 31 kilometres worldwide) for many different atmospheric variables, such as temperature, humidity, wind speed, precipitation, and surface pressure. In order to ensure accuracy and consistency, ERA5 integrates information from satellites, weather stations, and other sources using sophisticated numerical models and data assimilation techniques. It improves on ERA-Interim, its predecessor, by utilizing better model physics and higher-quality data. To maximize accuracy, configuration entails defining the model resolution, time intervals, parameterization schemes, and assimilation processes.

METHODOLOGY

Non hydrostatic advanced research weather research and forecasting (WRF-ARW) model version 4.1.2 is used for this study. The model is planned to focus on finer horizontal grid lengths of very less kilometres or less with nested domains for prediction and simulation of different good-scale

atmospheric phenomena (John et al., 2004; Skamarock et al., 2005). Eulerian mass dynamical core is used to consider the terrain hybrid sigma-pressure vertical coordinates. Runge Kutta 2nd and Runge Kutta 3rd order schemes are applied for time integration and 2nd to 6th order schemes are applied for advection. The model has various parametrization of physical processes, a data assimilation packages, and number of various physics schemes. It has various options for convective parameterization schemes, planetary boundary layer schemes, and microphysics schemes. The detailed model configuration of this study for the SuCS ‘Amphan’ simulation is given in Table 1.

Vertical integrated moisture flux Q can be expressed as

$$Q = \frac{1}{g} \int_0^{p_s} qV dp$$

where V is wind vector, q is specific humidity, p_s is surface pressure, and g is gravitational constant. For vertical integrated moisture transport (VIMT), vertical integration of this equation is carried out from surface pressure (in hPa) to 300 hPa since specific humidity have negligible effect above this level and is not a part of ERA5 reanalysis datasets (Kalnay et al., 1996).

Table 1. Model configuration used in WRF-ARW model

<i>Model Configuration</i>	<i>Configuration</i>
	<i>Bay of Bengal</i>
Dynamics	Nonhydrostatic
Model domain	4°N-26°N, 78°E-103°E
Domain center	15°N and 89.5° E
Horizontal resolution	9km
Map projection	Mercator
Grid scheme in horizontal	Arakawa C-grid
Time step	40s
Vertical coordinates	Hydrostatic pressure vertical coordinate with 51 pressure levels
Time integration schemes	3rd order R-K Scheme
Spatial differencing scheme	Center differencing (6th order)
Micro physics	Ferrari (5)
Radiation schemes (ra_lw_physics & ra_sw_physics)	RRTM Scheme for long wave/ Dudhiya for short wave
Surface layer parameterization (sf_surface_physics)	Thermal diffusion scheme
Planetary boundary layer physics (bl_pbl_physics)	Yonsei University (YSU)
Cumulus scheme (cu_physics)	Kain Fritsch Scheme

RESULTS AND DISCUSSION

The results obtained using advanced research weather research and forecasting (WRF-ARW) model with 9 km resolution in a single domain over Bay of Bengal (BoB) are discussed below. The parameters viz spatial distribution of wind, potential vorticity (PV), and vertical integrated

moisture transport (VIMT) are analysed and validated with fifth generation of European center for medium-range weather forecast (ECMWF) atmospheric reanalysis of global climate (ERA5) reanalysis datasets.

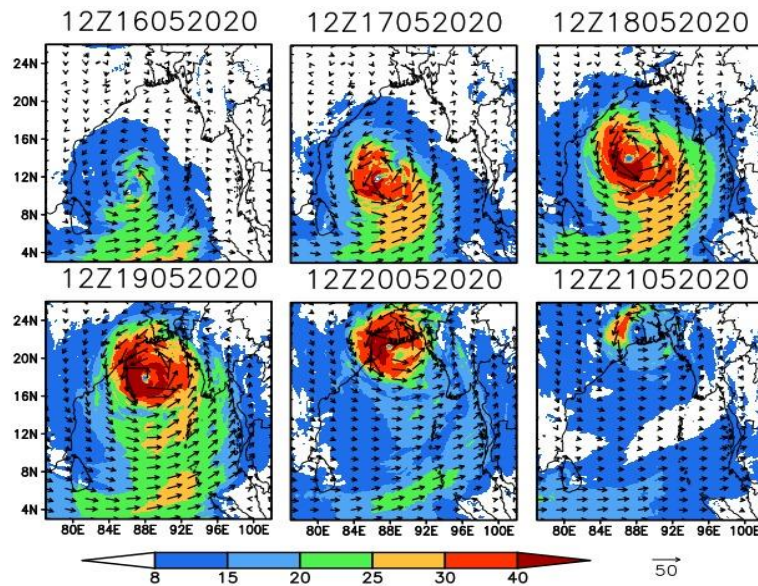


Figure 2. Pattern of wind flow (m/s) at 850 hPa from pre- to post- landfall (16 May 2020 - 21 May 2020) using advanced research weather research forecast (WRF-ARW) model experiment data.

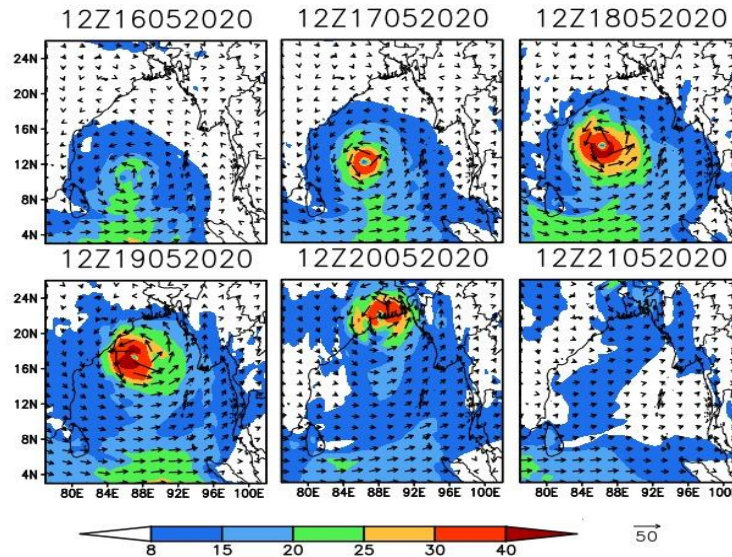


Figure 3. Pattern of wind flow (m/s) at 850 hPa from pre- to post- landfall (16 May 2020-21 May 2020) using ERA5 data.

Simulation of wind

Wind plays an important role in the dynamics of a tropical cyclone. Accurate wind pattern forecast is essential to minimise damage. Figure 2 depicts the low-level wind flow distribution (in m/s) at 850 hPa as simulated by the WRF-ARW model for SuCS ‘Amphan’. Figure 3 depicts the low-level wind at 850 hPa from ERA5 reanalysis datasets. In Figure 2, the model-simulated wind flow pattern, valid for 1200 UTC of May 16-21, 2020, based on the initial and boundary conditions at 1200 UTC on May 16, 2020. The model depicts a well-organized cyclonic circulation, characterized by strong winds encircling the center. Notably, the model accurately captured the maximum wind speed, reaching approximately 50 m/s, which closely matched with the ERA5 reanalysis datasets maximum wind speed. From all the figures, it is observed that the wind flows concentrated into DD of cyclone on 16th May 2020 1200 UTC, SCS converted into VSCS on 17th May 1200 UTC, VSCS converted into SuCS on 18th May 1200 UTC, SuCS weakening into an ESCS on 19th May noon, just before crossing west Bengal- Bangladesh coasts on 20th May 1200 UTC, and DD over Bangladesh on 21st May 2020.

It is observed from both data sets that the circulation of wind started on 16th May 2020 and moved towards west Bengal with higher magnitude of wind till 19th May 2020 and then started weakening around afternoon of 19th May. It can also be observed from Figure 2 that cyclone eye started developing on 17th May 2020 and started moving towards north eastwards. Cyclone eye is fully developed on 18th - 19th May 2020, with maximum speed around 50 m/s

and touched west Bengal nearby Kolkata. Cyclone eye keeps moving towards Bangladesh and started weakening during 20th-21st May. In this analysis, the pattern of wind observed from ERA5 re-analysis is quite similar to wind observed from model, but the intensity of wind is lower than the model wind intensity. WRF-ARW captures high speed of wind than ERA5. It may be because of low resolution of ERA5.

Potential vorticity analysis

The analysis of potential vorticity (PV) at 500 hPa from WRF-ARW and ERA5 reanalysis datasets during 16th May 2020 to 21st May 2020 is shown in Figures 4 and 5 respectively. The gradual increase of intensity of the PV is noticed from initial phase to established phase around inner region of the SuCS ‘Amphan’. The distribution of high value of PV (greater than and equals to 3.0 PVU where, 1PVU equals to $10^{-6} \text{ K} \cdot \text{m}^2 \cdot \text{kg}^{-1} \cdot \text{s}^{-1}$) is captured by WRF-ARW model in SuCS ‘Amphan’ inner region. The maximum PV region intensified the movement of cyclonic structure. At the same time, near the southwest portion of the cyclonic disorder, the low PV is noticed. This dynamical discrepancy provides a perfect condition for a powerful moisture fluxes carriage (Figure 6) from domain region into the internal storm’s core neighbourhood. The analysis showed visible band type shaped PV near the storm. Therefore, potential vorticity (PV) value rises till 19th May 2020 afternoon. Similar pattern of PV is also foreseen near the SuCS ‘Amphan’ by the ERA5 reanalysis datasets during cyclone cycle, although, the intensity of this pattern is low in ERA5 reanalysis datasets.

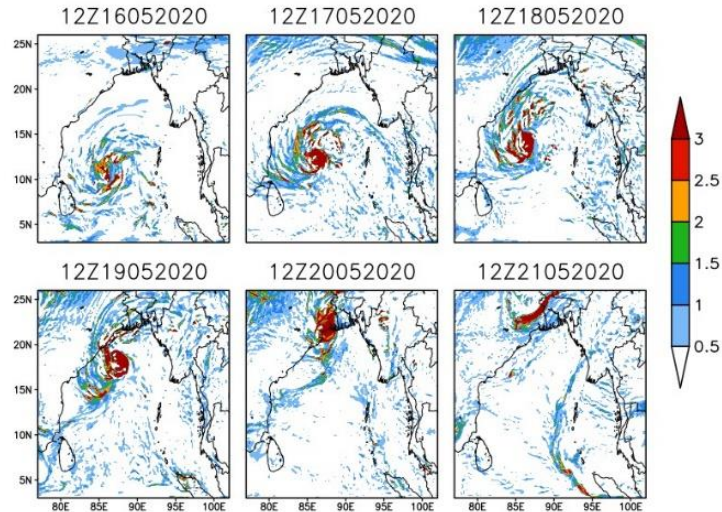


Figure 4. Potential vorticity at 500 hPa from pre- to post- landfall (16 May 2020–21 May 2020) using advanced research weather research forecast (WRF-ARW) model experiment data.

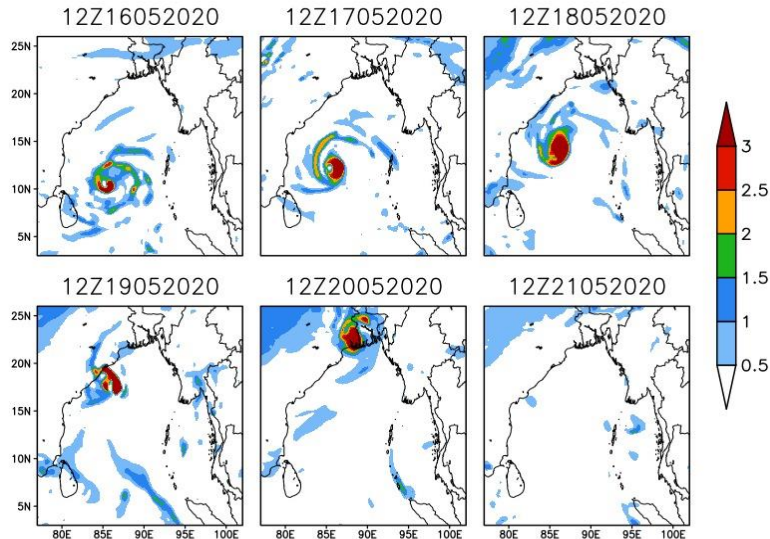


Figure 5. Potential vorticity at 500 hPa from pre- to post- landfall (16 May 2020–21 May 2020) using ERA5 data.

Vertical integrated moisture transport (VIMT)

Figures 6 and 7 depict the spatial spreading of vertical integrated moisture transport (VIMT) and VIMT magnitude ($\text{kg. m}^{-1} \cdot \text{s}^{-1}$) from WRF-ARW output and ERA5 reanalysis datasets during 16th May 2020 to 21st May 2020. A wide role in the tropical cyclone (TC)'s development and intensity is contributed by the latent heat (Hill and Lackmann, 2009; Huang et al., 2014; Makarieva et al., 2017; Fujiwara et al., 2017).

Latent heat is often transferred by moisture transport in the SuCS 'Amphan' system. In WRF-ARW on 16th May at 1200 UTC (Figure 7), at the initial phase of the tropical

cyclone (TC) system, the moist northeast flowing around the Indian ocean is clearly observed. From 16th May 2020 1200 UTC to 19th May 2020 1200 UTC (Figure 7), the eastward transportation of humidity is progressive over Bay of Bengal (BoB) north east region. The moisture in north-eastward is constantly mixing near SuCS 'Amphan' over domain region. Hence, the system is well connected with the increased moisture so that this increment in moisture results a moisture conveyor belt (MCB) formation over the region. A well-marked low pressure (WML) system was established on 17th May 1200 UTC as reported by ERA5 reanalysis datasets.

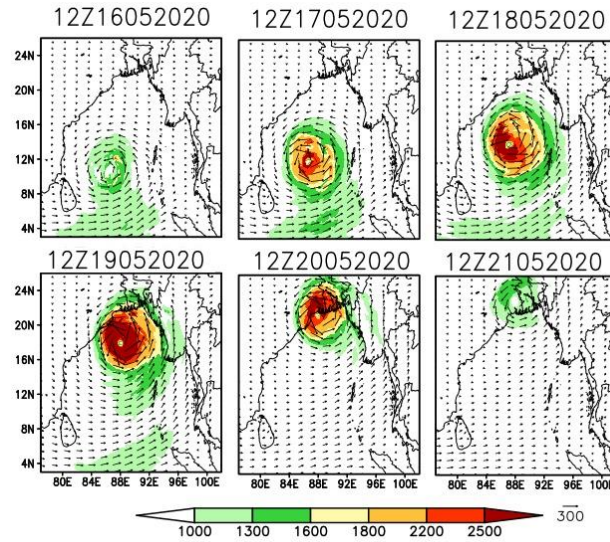


Figure 6. Vertical integrated moisture transport (VIMT) at different stages from pre- to post- landfall (16 May 2020–21 May 2020) using advanced research weather research forecast (WRF-ARW) model experiment data.

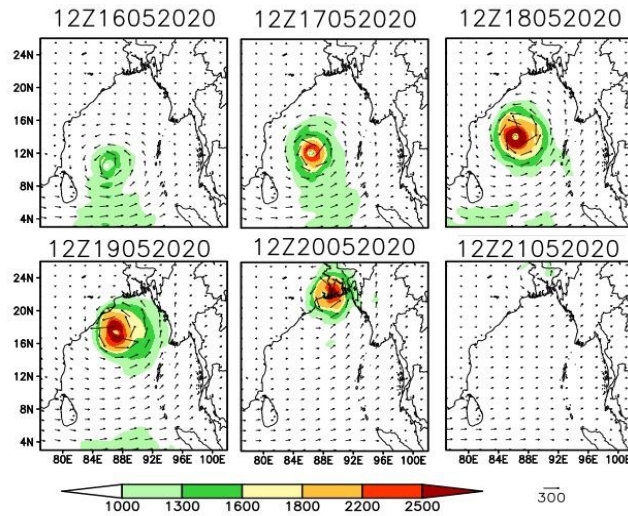


Figure 7. Vertical integrated moisture transport (VIMT) at different stages from pre- to post- landfall (16 May 2020–21 May 2020) using ERA5 data.

In the Figure 7, on 19th May, 2020 1200 UTC the WML system is observed. The intensity of storm from DD to CS started increasing during the period 16th to 17th May, 2020 (Figure 7) and moved north-eastward over the northeast Bay of Bengal (BoB) adjoining Bangladesh. The cyclonic storm (CS) moved over the north-eastward region. In the inner area of the cyclonic arrangement, there is a large-scale transportation of moisture from the ocean by moisture conveyor belt (MCB). This large-scale moisture transport increased the intensity of the tropical cyclones (TCs) as

The shape and development of the SuCS ‘Amphan’ are also observed on 17th -18th May, 2020 and both are also very clearly matched with ERA5 reanalysis datasets. The model

more than $550 \text{ kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$ absolute value of moisture fluxes is taken back by the moisture conveyor belt (Fujiwara et al., 2017; Kudo et al., 2014; Takakura et al., 2018). The progress of the SuCS ‘Amphan’ is steady and shown in all the model analysis (Figure 6). The absolute value of VIMT and MCB pattern of WRF-ARW model are significantly matched with the ERA5 reanalysis (Figure 7; 18th – 20th, 2020). The model forecasts are successfully reproduced above mentioned characteristics observed in the ERA5 reanalysis datasets.

predicted moisture conveyor belt (MCB) time evolution considerably fine. Hegde et al. (2016), Fujiwara et al. (2017) and Makarieva et al. (2017) simulated similar

features of VIMT through regional atmospheric models over South China sea and Indian ocean. The moisture conveyor belt (MCB) development is an important aspect for the moisture import inside system core from the remote oceans. This development of moisture conveyor belt (MCB) subsequently increases the latent heat which cause the increase in storm intensity. The storm intensification progressed consequent to additional moisture conveyor belt (MCB) development. The absolute value of VIMT of the SuCS 'Amphan' is more than $950 \text{ kg. m}^{-1}.\text{s}^{-1}$ and it is higher than the ERA5 reanalysis datasets over SuCS 'Amphan' surrounding and inner region. Therefore, the time evolution of large-scale vapour transportation is predicted well by the high resolution WRF-ARW model during SuCS 'Amphan' initial stage.

CONCLUSIONS

This study investigated the predictive ability in terms of different parameters viz, spatial distribution of wind, potential vorticity (PV), and vertical integrated moisture transport (VIMT) of advanced research weather research and forecasting (WRF-ARW) model, during the developing features connected with the super cyclonic storm (SuCS) 'Amphan' over BoB from 16th – 21st May, 2020. The wind field using model simulation at 850 hPa shows that cyclonic rotation increases with the increase in simulation lengths. The model simulated wind field shape is quite matched with the ERA5 re-analysis shape during the simulation length. The intensity of model simulated potential vorticity (PV) growing around the inner part of super cyclonic storm (SuCS) 'Amphan' from initial stage of SuCS 'Amphan' to intense stage of SuCS 'Amphan'. The small value of PV is observed in the north east part of the SuCS 'Amphan'. The storm's features are well matched with ERA5 datasets with high PV values, and it keeps tropical cyclone (TC) initial strength and structure during the evolved stage. We also observed response between intensity of the super cyclonic storm (SuCS) 'Amphan' during pre-landfall and moisture transportation over the region of model through vertical integrated moisture transport (VIMT) distribution. It is found from WRF-ARW model results that the moist flows over BoB region are increased with formation of SuCS 'Amphan' and moisture conveyor belt (MCB), formed due to these moist flows over model domain. The intensity of the SuCS 'Amphan' increased due to big scale transportation of moisture inside region of cyclone from the ocean through moisture conveyor belt (MCB). The moisture conveyor belt (MCB) pattern and magnitude of VIMT over Indian ocean are almost correlated with ERA5 reanalysis datasets. The

results obtained from WRF-ARW model are in agreement with the previous results from ERA5 reanalysis datasets. It is concluded that the advanced research weather research and forecasting (WRF-ARW) model, can predict all stages of the super cyclonic storm (SuCS) 'Amphan.' Simulations of more tropical cyclones (TCs) with sensitivity analysis and the obtained results, required validation with different source of observations for the support of the conclusions of the present study.

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Author Credit Statement

Mr. Arun Kumar, and Dr. Sushil Kumar carried out all the experiments, conceptualise it and designed the study, besides data acquisition, analysis, interpretation of data and writing of the paper. Prof. Nagendra Kumar substantially contributed in conceptualisation and revision of the manuscript. Mr. Nitin Lohan customized the model and analysed the results., Prof. Rajeev Bhatla analysed the results.

Data Availability

ERA5 is a gridded hourly data which is available on pressure level at

<https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-pressure-levels?tab=overview> and ERA5 hourly data on single level is available on

<https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels?tab=overview>. In this research, access to NCEP data was facilitated through the following site <https://rda.ucar.edu/datasets/ds351.0/dataaccess/>.

Compliance with Ethical Standards

The authors declare that they have no conflict of interest and adhere to copy right norms.

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